

Features

- **Services:** Messaging, clustering, containers, compute, identity, app data protection as a service, events, metadata indexing as service, workflows, DNS, database as a service, bare metal provisioning, optimisation and deployment, governance, and benchmarking.
- Web front-end, Big Data processing framework, container orchestration engine, key management, and NFV orchestration.

Official website: <http://www.openstack.org>

Latest version: Stein

ManageIQ

This is an open source CMP for hybrid IT environments, with a mix of public and private clouds. It provides tools for managing small and large environments as well as supports multiple technologies like virtual machines, public clouds and containers. It allows users to download any virtual appliance and deploy copies of it into virtualisation platforms like OpenStack or VMware. Three main variants of ManageIQ are available: Vagrant, Docker and Public Cloud.

ManageIQ is written in Ruby and uses the Ruby on Rails framework. The ManageIQ software is shipped as a pre-built virtual appliance, roughly 1GB in size. The appliance is based on the CentOS operating system and includes an embedded PostgreSQL database. Since the Darga release, a container based version has also been made available. An appliance can be used on its own, or it can be part of a three-tier federated architecture. In the latter case, the operator configures zones, regions and a single super-region. Appliances can be assigned to a specific zone or region and are configured with specific roles so that work is coordinated within the region. Most roles are multi-master and distribute work automatically in a

queue, but some roles like the database are singletons.

Features

- Offers insights through discovery, monitoring, utilisation, performance, reporting, analytics, chargeback and trending.
- Controls security, compliance, alerting, policy-based resource and configuration management.
- Automates IT processes, tasks and events, provisioning, workload management and orchestration.
- Integrates systems management, tools and processes, event consoles, CMDB, RBA and Web services.

Official website: <http://manageiq.org>

Latest version: Hammer-10

Cloudify

Cloudify is an open source software cloud and NFV orchestration product that uses OASIS TOSCA technology. It is designed using Python. Cloudify allows users to model and automate an application's entire life cycle. This includes deployment to a cloud or data centre environment, the management of the deployed application, failure detection and ongoing maintenance. The platform is ideal for users who want to launch prebuilt applications in the cloud without handling the technical aspects.

How it works: It translates applications into a blueprint configuration written in the YAML format and describes how the application should be deployed, managed and automated. It identifies the resources and events for every application tier. The cloud orchestrator uses blueprints to install applications in the cloud using a cloud API, which creates VMs and installs Cloudify agents, and is used to orchestrate, install and start the application. Cloudify monitors the application for any pre-defined metrics and displays results on the dashboard.

Cloudify enables users to deploy applications using two main methods — by using the CLI and by using a Cloudify manager. The latter is a dedicated environment comprising an open source stack which enables the user to:

- Use plugins (such as Docker, Script, Chef and Puppet plugins) to manage application hosts.
 - Keep a directory of the user's blueprints.
 - Create multiple deployments for each blueprint and install them.
 - Execute healing, scaling and other custom workflows on installed applications.
 - Run multiple workflows concurrently.
 - View an application's topology.
 - Perform different tasks using the Cloudify Web UI view metrics.
- Cloudify performs the following tasks.

- **Application modelling:** This describes the application with all its resources.
- **Orchestration:** This maintains and runs an application, and performs ongoing operations such as scaling, healing and maintenance.
- **Pluggability:** This provides reusable component abstraction for the system.
- **Security:** This provides secure communication via SSL, which enables clients to ensure that the data set received is encrypted.

Features

Easy orchestration, built-in node types, a blueprints catalogue, role-based access control, IT security and governance, network and TOSCA orchestration, new NFV capabilities, custom widgets and LDAP integration.

Official website: <http://www.cloudify.co>

Latest version: 5.0

Mist.io

Mist.io is a platform that simplifies